

Epistemic Information Gain, Entropy, and Entropy Collapse in Reinforcement Learning

Abstract

This note gives a first-principles account of how to measure information gained by an RL agent after learning from a rollout on a single environment. The core claim is simple: entropy reduction is meaningful only after naming the uncertainty object whose entropy is being reduced. Entropy over actions measures how decisive a policy is. Entropy over latent hypotheses measures epistemic uncertainty. Expected epistemic information gain is expected posterior entropy reduction, equivalently mutual information. In practical RL systems, this ideal quantity is measured differently for a single non-Bayesian policy, an ensemble of policies, and a Bayesian policy. The document ends with concrete estimators and diagnostics for separating useful information gain from entropy collapse.

1 The First Principle: Information Is Uncertainty Reduction

Let Z be an unknown quantity. It could be a hidden task identity, a reward parameter, a transition parameter, a value-function hypothesis, or a policy parameter. A belief over Z is a probability distribution $p(z)$. The entropy of that belief is

$$H[Z] = - \sum_z p(z) \log p(z), \quad (1)$$

for a discrete variable. Entropy is high when the belief mass is spread across many alternatives and low when the belief is concentrated.

If a new observation Y is collected, the belief changes from $p(z)$ to $p(z | y)$. The information gained from the observation is the reduction in uncertainty:

$$H[Z] - H[Z | Y = y]. \quad (2)$$

Before observing Y , the possible outcomes are unknown, so the expected information gain is

$$I(Z; Y) = H[Z] - \mathbb{E}_{y \sim p(y)} H[Z | Y = y]. \quad (3)$$

This is mutual information. An equivalent expression is an expected KL divergence:

$$I(Z; Y) = \mathbb{E}_{y \sim p(y)} \text{KL}(p(z | y) \parallel p(z)). \quad (4)$$

This equivalence is important in RL. The entropy-reduction view says “uncertainty went down.” The KL view says “the observation changed the belief.” For continuous latent variables, differential entropy can be coordinate-sensitive, so the KL form is often the cleaner measurement.

2 Epistemic Information Gain From One Environment

Let $\mathcal{D}$ be the agent’s current data and let $e \in \mathcal{E}$ be a particular environment or level. A rollout from that environment is

$$\tau = (s_0, a_0, r_0, s_1, \dots, s_T). \quad (5)$$

Let θ denote the latent object the agent is uncertain about. The exact choice of θ depends on the scientific question:

- transition dynamics parameters;
- reward parameters;
- task identity;
- value-function hypotheses;
- policy parameters;
- the identity of which action is optimal in each state.

The expected epistemic information gain of rolling out on environment e is

$$\text{EIG}(e) = H[\theta \mid \mathcal{D}] - \mathbb{E}_{\tau \sim p(\tau|e, \mathcal{D})} H[\theta \mid \mathcal{D}, \tau]. \quad (6)$$

Equivalently,

$$\text{EIG}(e) = I(\theta; \tau \mid \mathcal{D}, e) = \mathbb{E}_{\tau \sim p(\tau|e, \mathcal{D})} \text{KL}(p(\theta \mid \mathcal{D}, \tau) \parallel p(\theta \mid \mathcal{D})). \quad (7)$$

This is the clean target. It is epistemic because the entropy is over what is true but unknown, not over stochastic actions or random outcomes. It is expected because the rollout has not yet happened when choosing which environment to sample.

3 Nearby Methods for Scoring Informative Environments

Several recent methods are close to the question in this note, but they place the information signal in different objects: policy parameters, learned dynamics, ensemble predictions, or task-policy relationships. This section summarizes four representative methods and relates each to the ideal quantity $I(\theta; \tau \mid \mathcal{D}, e)$.

3.1 PACE: Parameter Change as Realized Learning Progress

PACE, proposed by Yuan et al., is the closest method to the single-policy setting. It starts from the observation that many UED scores are indirect: regret requires an optimal policy or an expert, value-error scores depend on the critic, and Monte Carlo scores can have high variance. PACE instead scores a level by the policy parameter change induced by training on that level.

Let θ^- be the policy parameters before a provisional update on level e , and let θ_e^+ be the parameters after that local update. The PACE score is essentially

$$S_{\text{PACE}}(e) = \left\| \theta_e^+ - \theta^- \right\|_2^2. \quad (8)$$

The motivation is a first-order Taylor approximation. If a local gradient update on level e is aligned with the level-specific objective, then the objective improvement induced by that level is

proportional to the squared norm of the corresponding parameter update, up to a step-size constant. PACE therefore treats parameter movement as an intrinsic, low-variance proxy for realized learning progress.

Operationally, PACE collects a trajectory on a candidate level, performs a temporary update to compute the induced parameter change, inserts high-scoring levels into a replay buffer, and samples from that buffer with prioritization and staleness correction. This maps naturally to the single non-Bayesian policy regime in this note: it is not epistemic IG, because there is no posterior, but it is a direct approximation to “this environment caused the current learner to move.”

3.2 MaxInfoRL: Information Gain as an Intrinsic Reward

MaxInfoRL, proposed by Sukhija et al., uses information gain inside online RL exploration rather than as an external environment-selection score. The agent learns a model of the unknown transition-and-reward function f^* and uses an information-gain bonus to direct exploration toward transitions that reduce uncertainty about f^* .

For a transition query (s, a) , the ideal intrinsic reward is of the form

$$r_{\text{IG}}(s, a) = \text{I}(\tilde{s}', f^* \mid s, a, \mathcal{D}), \quad (9)$$

where $\tilde{s}'$ denotes the predicted next transition target, possibly including reward. In practice, the information gain does not generally have a closed form, so MaxInfoRL uses an upper bound derived from epistemic uncertainty in a learned model. With an ensemble or probabilistic model, the practical bonus is high where model disagreement about the transition or reward is high.

MaxInfoRL then combines this intrinsic reward with maximum-entropy actor-critic updates. In soft actor-critic notation, the policy is no longer pushed only toward high extrinsic value and action entropy; it is also pushed toward states and actions where the model expects high information gain. This method is a true information-gain method with respect to dynamics/reward uncertainty, but its unit of selection is a transition or behavior policy, not necessarily a reusable environment level.

3.3 MAX: Ensemble Disagreement for Active Exploration

Model-Based Active Exploration (MAX), proposed by Shyam, Jaskowski, and Gomez, also treats exploration as an active information-gathering problem. MAX trains an ensemble of forward dynamics models and plans in a surrogate exploration MDP whose reward is novelty of predicted transitions.

For a discrete next-state distribution, the ideal ensemble disagreement score is

$$u_{\text{MAX}}(s, a) = \text{H} \left[\frac{1}{N} \sum_{i=1}^N p_i(s' \mid s, a) \right] - \frac{1}{N} \sum_{i=1}^N \text{H}[p_i(s' \mid s, a)]. \quad (10)$$

This is the same entropy-of-the-mean minus mean-entropy structure used earlier for action-space epistemic uncertainty. Here, however, it is applied to next state predictions rather than action distributions. In continuous domains, MAX replaces Shannon entropy with Renyi entropy and uses a Jensen-Renyi divergence so that disagreement among Gaussian ensemble predictions can be computed tractably.

MAX is therefore an ensemble approximation to epistemic information gain about the dynamics. It is especially relevant to the ensemble-policy section of this note because it shows the same mathematical decomposition in a different space: not “which action do policy hypotheses prefer?” but “which transition do dynamics hypotheses predict?”

3.4 ITTS: Information-Theoretic Task Selection for Meta-RL

Information-Theoretic Task Selection (ITTS), proposed by Luna Gutierrez and Leonetti, studies which training tasks should be selected for meta-RL. Unlike PACE, MaxInfoRL, and MAX, ITTS is not primarily about online exploration within a single environment. It asks which tasks are useful members of a meta-training set.

ITTS keeps a candidate training set and a validation task set. It selects source tasks that are both different from already selected tasks and relevant to the validation tasks. Difference is measured by average KL divergence between the policies learned for two candidate tasks, evaluated over states sampled from the validation tasks:

$$\delta(m_1, m_2) = \frac{1}{|\mathcal{S}_{\text{ref}}|} \sum_{s \in \mathcal{S}_{\text{ref}}} \text{KL}(\pi_{m_1}^*(\cdot | s) \| \pi_{m_2}^*(\cdot | s)). \quad (11)$$

Relevance is measured by adapting a policy from one task to another and checking how the policy entropy changes over the validation distribution. A task is useful when it is not redundant with selected tasks and transfers information to the validation tasks.

For the present note, ITTS is important because it treats information as a relationship between tasks mediated by policies. It is less directly applicable to scoring a single rollout on one environment, but it suggests a useful evaluation question: does learning on environment e_i change the policy in a way that transfers to a target task distribution, or only in a way that is locally decisive on e_i ?

3.5 What Has Been Compared Carefully?

The clearest metric-comparison work I found is No Regrets by Rutherford et al. It asks whether common UED prioritization scores actually approximate the quantities they are meant to approximate. In domains where regret can be computed or probed more directly, they compare practical scores such as MaxMC and positive value loss against regret, success rate, and learnability. Their key finding is that practical regret approximations can correlate with success rate rather than learnability, causing replay to spend experience on levels the agent has already mastered.

PACE also compares against UED baselines such as DR, PLR, robust PLR, and ACCEL, and explicitly discusses bias, variance, and computational cost of value-based, Monte Carlo, regret, and marginal-benefit signals. However, it is primarily an algorithmic comparison, not a full oracle study of every approximation against a shared true information-gain target.

MaxInfoRL and MAX compare information-gain or disagreement-based exploration against exploration baselines. ITTS compares selected task subsets against using all or other subsets of meta-training tasks. I did not find a unified benchmark that places PACE-style parameter change, MaxInfoRL-style model information gain, MAX-style ensemble disagreement, and ITTS-style task-policy transfer under one oracle information-gain evaluation.

4 Entropy in RL

RL contains several different entropies. Confusing them is the main source of bad information-gain measurements.

4.1 Policy Action Entropy

For a stochastic policy $\pi(a | s)$, action entropy at state s is

$$H_\pi(s) = - \sum_{a \in \mathcal{A}} \pi(a | s) \log \pi(a | s). \quad (12)$$

This measures how spread out the policy’s action distribution is. Low action entropy means the policy is decisive. High action entropy means the policy is diffuse or exploratory.

For a fixed probe state set $\mathcal{S}_{\text{ref}}$, the average action entropy is

$$\bar{H}_{\pi}(\mathcal{S}_{\text{ref}}) = \frac{1}{|\mathcal{S}_{\text{ref}}|} \sum_{s \in \mathcal{S}_{\text{ref}}} H_{\pi}(s). \quad (13)$$

4.2 Trajectory Entropy

A policy and environment together induce a distribution over trajectories:

$$p(\tau \mid e, \pi). \quad (14)$$

The entropy of this distribution measures how many possible trajectories the agent-environment system can generate. It mixes policy stochasticity, transition stochasticity, and state visitation diversity.

4.3 Return Entropy

The entropy of returns measures variability in cumulative reward:

$$G(\tau) = \sum_{t=0}^T \gamma^t r_t. \quad (15)$$

This is often about outcome uncertainty rather than epistemic uncertainty. A fully understood environment can still have high return entropy if it is intrinsically stochastic.

4.4 Belief Entropy

Belief entropy is entropy over latent hypotheses:

$$H[\theta \mid \mathcal{D}]. \quad (16)$$

This is the entropy that directly supports epistemic information gain. In a Bayesian model it is explicit. In an ensemble it is approximated. In a single non-Bayesian policy it is usually absent, so proxy metrics are required.

5 Decisiveness Is Not Epistemic Information

Suppose a policy has two actions at state s :

$$\pi_{\text{before}}(\cdot \mid s) = (0.5, 0.5), \quad \pi_{\text{after}}(\cdot \mid s) = (0.95, 0.05). \quad (17)$$

The action entropy dropped. The agent became more decisive. But this does not prove that the agent learned the truth about the environment. The same pattern can arise from:

- correct learning from informative data;
- a changed entropy coefficient or temperature;
- optimizer instability;

- overfitting to a noisy rollout;
- a policy update that sharpened preferences without reducing model uncertainty.

The distinction is:

$$\text{Decisiveness} = \text{concentration of } \pi(a \mid s), \quad (18)$$

$$\text{Epistemic information} = \text{reduction of uncertainty about } \theta. \quad (19)$$

Action entropy reduction is useful, but it is not automatically epistemic.

6 Entropy Collapse in RL

Entropy collapse is the rapid reduction of policy action entropy before the agent has learned a robust, generalizable decision rule. It is not just “becoming confident.” It is becoming confident faster than the evidence supports.

6.1 Mechanisms

Common mechanisms include:

- a small or decaying entropy bonus in policy-gradient training;
- large policy updates that overreact to high-variance advantages;
- sparse rewards that reinforce a lucky early trajectory;
- replay systems that repeatedly sample narrow states or levels;
- clipping or normalization choices that hide value-function uncertainty;
- deterministic action selection during data collection.

6.2 Why Entropy Collapse Matters

Entropy collapse can masquerade as information gain. If

$$\bar{H}_{\pi_{\text{before}}}(\mathcal{S}_{\text{ref}}) - \bar{H}_{\pi_{\text{after}}}(\mathcal{S}_{\text{ref}}) \quad (20)$$

is large, the policy became more concentrated. But that number alone cannot say whether the rollout was informative. A bad rollout can make the agent confident in the wrong action. In that case, entropy reduction is a warning sign, not a success metric.

6.3 Diagnostics

Useful diagnostics for entropy collapse are:

- action entropy reduction on a fixed probe set;
- return or success-rate improvement on held-out environments;
- policy KL step size before and after learning;
- value-loss change on held-out trajectories;

- ensemble or posterior disagreement, when available;
- calibration of predicted values against realized returns.

A healthy update has a coherent pattern: the policy changes, held-out loss or return improves, and epistemic disagreement decreases. An unhealthy collapse has policy sharpening without corresponding held-out improvement or disagreement reduction.

7 Measuring After Learning on a Single Environment

The experimental unit is one environment e . The measurement protocol should compare the agent before and after learning from rollouts on e while holding the evaluation states fixed.

7.1 Protocol

1. Save the pre-learning agent state.
2. Choose a fixed reference state set $\mathcal{S}_{\text{ref}}$. This may be collected from a broad replay buffer, held-out levels, random starts, or a standardized evaluation distribution.
3. Sample K independent rollout batches $\tau_1, \dots, \tau_K \sim p(\tau \mid e, \mathcal{D})$.
4. For each rollout batch τ_k , copy the pre-learning agent and run the same learning procedure on $\mathcal{D} \cup \tau_k$.
5. Measure before and after quantities on the same $\mathcal{S}_{\text{ref}}$.
6. Average the after-learning quantities over k .

The fixed $\mathcal{S}_{\text{ref}}$ matters. If entropy is measured only on states visited by each rollout, the metric confounds information gain with a changed state distribution.

8 Single Non-Bayesian Policy

A single non-Bayesian policy has parameters θ but no posterior $p(\theta \mid \mathcal{D})$. Therefore direct epistemic information gain is not available. The practical target becomes learning progress: did data from environment e cause a useful, generalizing update?

8.1 Action Entropy Reduction

For rollout batch τ_k , let π_k^+ be the policy after learning on τ_k and π^- the policy before learning. Define

$$\Delta H_{\text{action}}(e) = \bar{H}_{\pi^-}(\mathcal{S}_{\text{ref}}) - \frac{1}{K} \sum_{k=1}^K \bar{H}_{\pi_k^+}(\mathcal{S}_{\text{ref}}). \quad (21)$$

This measures decisiveness gain. It is not epistemic information gain by itself.

8.2 Policy Update Magnitude

A direct measure of how much the rollout changed the policy is

$$\Delta_{\text{KL}}(e) = \frac{1}{K} \sum_{k=1}^K \frac{1}{|\mathcal{S}_{\text{ref}}|} \sum_{s \in \mathcal{S}_{\text{ref}}} \text{KL}(\pi_k^+(\cdot | s) \| \pi^-(\cdot | s)). \quad (22)$$

A symmetric version may be used when directional KL is unstable:

$$\Delta_{\text{SKL}}(e) = \frac{1}{2} \text{KL}(\pi_k^+ \| \pi^-) + \frac{1}{2} \text{KL}(\pi^- \| \pi_k^+). \quad (23)$$

8.3 Probe Loss Reduction

Let $\mathcal{D}_{\text{probe}}$ be a fixed held-out dataset of states, actions, returns, advantages, or value targets. Let $L(\theta; \mathcal{D}_{\text{probe}})$ be a chosen loss, such as PPO surrogate loss, behavior-cloning loss, TD error, value loss, or return-prediction loss. Define

$$\Delta L_{\text{probe}}(e) = L(\theta^-; \mathcal{D}_{\text{probe}}) - \frac{1}{K} \sum_{k=1}^K L(\theta_k^+; \mathcal{D}_{\text{probe}}). \quad (24)$$

This is one of the best single-policy proxies because it asks whether learning from e improved performance on data not identical to the update rollout.

8.4 Rollout Surprise

If the agent has value predictions, define TD error

$$\delta_t = r_t + \gamma V_{\theta^-}(s_{t+1}) - V_{\theta^-}(s_t). \quad (25)$$

Then

$$S_{\text{TD}}(e) = \frac{1}{K} \sum_{k=1}^K \sum_{t \in \tau_k} |\delta_t| \quad (26)$$

measures how surprising the rollout was under the current value function. High surprise can identify candidate informative environments, but surprise without subsequent probe improvement may be noise rather than information.

8.5 Recommended Single-Policy Panel

For a single non-Bayesian policy, report:

$$(\Delta H_{\text{action}}, \Delta_{\text{KL}}, \Delta L_{\text{probe}}, S_{\text{TD}}, \Delta \text{return}_{\text{heldout}}). \quad (27)$$

Interpretation should be conservative:

- entropy reduction alone means decisiveness;
- policy KL means the rollout changed the policy;
- probe loss reduction means the update was useful;
- TD error means the rollout contradicted the critic;
- held-out return improvement is the final behavioral check.

9 Ensemble Policy

With N policies, each policy can be treated as a hypothesis:

$$\pi_1, \dots, \pi_N. \quad (28)$$

These may come from different random seeds, bootstrap resampling, different checkpoints, or different training perturbations. This is not automatically Bayesian, but it gives an empirical approximation to epistemic uncertainty.

9.1 Predictive Action Distribution

For each state s , define the ensemble-average action distribution:

$$\bar{\pi}(a \mid s) = \frac{1}{N} \sum_{i=1}^N \pi_i(a \mid s). \quad (29)$$

The total predictive action entropy is

$$\mathbf{H}[\bar{\pi}(\cdot \mid s)]. \quad (30)$$

The average individual policy entropy is

$$\frac{1}{N} \sum_{i=1}^N \mathbf{H}[\pi_i(\cdot \mid s)]. \quad (31)$$

9.2 Disagreement as Epistemic Uncertainty

The ensemble disagreement term is

$$U_{\text{epi}}(s) = \mathbf{H}[\bar{\pi}(\cdot \mid s)] - \frac{1}{N} \sum_{i=1}^N \mathbf{H}[\pi_i(\cdot \mid s)]. \quad (32)$$

This is high when individual policies are confident but disagree. It is low when the policies agree, even if each policy is stochastic.

Average over the probe states:

$$U_{\text{epi}}(\mathcal{S}_{\text{ref}}) = \frac{1}{|\mathcal{S}_{\text{ref}}|} \sum_{s \in \mathcal{S}_{\text{ref}}} U_{\text{epi}}(s). \quad (33)$$

9.3 Information-Gain Proxy After Learning on One Environment

Let the ensemble before learning be $\{\pi_i^-\}_{i=1}^N$. For rollout batch τ_k , train or update the ensemble to obtain $\{\pi_{i,k}^+\}_{i=1}^N$. The ensemble epistemic-gain proxy is

$$\widehat{\text{EIG}}_{\text{ens}}(e) = U_{\text{epi}}^-(\mathcal{S}_{\text{ref}}) - \frac{1}{K} \sum_{k=1}^K U_{\text{epi},k}^+(\mathcal{S}_{\text{ref}}). \quad (34)$$

Separately, report decisiveness gain:

$$\Delta \mathbf{H}_{\text{individual}}(e) = \frac{1}{|\mathcal{S}_{\text{ref}}|} \sum_{s \in \mathcal{S}_{\text{ref}}} \left[\frac{1}{N} \sum_i \mathbf{H}[\pi_i^-(\cdot \mid s)] - \frac{1}{K} \sum_k \frac{1}{N} \sum_i \mathbf{H}[\pi_{i,k}^+(\cdot \mid s)] \right]. \quad (35)$$

The separation is essential:

- drop in individual entropy means the policies became sharper;
- drop in disagreement means the policies became more aligned;
- only the second is an epistemic proxy.

10 Bayesian Policy

A Bayesian policy maintains a posterior over parameters or hypotheses:

$$p(\theta \mid \mathcal{D}). \quad (36)$$

There may be one deployed agent, but its prediction is posterior predictive:

$$p(a \mid s, \mathcal{D}) = \mathbb{E}_{\theta \sim p(\theta \mid \mathcal{D})} \pi_{\theta}(a \mid s). \quad (37)$$

10.1 Action-Space Epistemic Uncertainty

At a probe state s , total predictive action entropy is

$$\mathbb{H} [\mathbb{E}_{\theta} \pi_{\theta}(\cdot \mid s)]. \quad (38)$$

The expected policy stochasticity is

$$\mathbb{E}_{\theta} \mathbb{H} [\pi_{\theta}(\cdot \mid s)]. \quad (39)$$

The epistemic action uncertainty is the difference:

$$U_{\text{Bayes}}(s) = \mathbb{H} [\mathbb{E}_{\theta} \pi_{\theta}(\cdot \mid s)] - \mathbb{E}_{\theta} \mathbb{H} [\pi_{\theta}(\cdot \mid s)]. \quad (40)$$

This is the mutual information between the action and posterior hypothesis:

$$U_{\text{Bayes}}(s) = \mathbb{I}(a; \theta \mid s, \mathcal{D}). \quad (41)$$

10.2 Posterior Update From One Environment

After rollout τ_k on environment e , the posterior becomes

$$p(\theta \mid \mathcal{D}, \tau_k). \quad (42)$$

The action-space epistemic information-gain estimator is

$$\widehat{\text{EIG}}_{\text{Bayes-action}}(e) = U_{\text{Bayes}}^{-}(\mathcal{S}_{\text{ref}}) - \frac{1}{K} \sum_{k=1}^K U_{\text{Bayes},k}^{+}(\mathcal{S}_{\text{ref}}). \quad (43)$$

This is an action-space proxy for epistemic information gain. The more direct Bayesian quantity is the posterior KL:

$$\widehat{\text{EIG}}_{\text{Bayes}}(e) = \frac{1}{K} \sum_{k=1}^K \text{KL} (p(\theta \mid \mathcal{D}, \tau_k) \parallel p(\theta \mid \mathcal{D})). \quad (44)$$

When the posterior is represented by samples $\theta_1, \dots, \theta_N$, the computation resembles the ensemble estimator. The interpretation is stronger: the samples approximate a posterior, not merely a collection of independently trained policies.

11 Expected Entropy Reduction Over Action Space

If the desired metric is explicitly expected entropy reduction over actions, the estimator is

$$\text{EER}_{\text{action}}(e) = \bar{H}_{\pi^-}(\mathcal{S}_{\text{ref}}) - \frac{1}{K} \sum_{k=1}^K \bar{H}_{\pi_k^+}(\mathcal{S}_{\text{ref}}). \quad (45)$$

This is valid as a measure of action-space entropy reduction. It should be named as such. It becomes an epistemic metric only when the action distribution is posterior predictive and the epistemic component is separated from expected policy stochasticity.

12 Practical Measurement Table

Setting	Main estimator	Interpretation
Single-policy, non-Bayesian	$\Delta L_{\text{probe}}, \Delta \text{KL}, \Delta H_{\text{action}}, \text{TD error}$	Learning-progress proxy. Entropy reduction means decisiveness, not direct epistemic information gain.
Ensemble policy	$U_{\text{epi}} = H[\bar{\pi}] - \frac{1}{N} \sum_i H[\pi_i]$	Disagreement reduction approximates epistemic uncertainty reduction.
Bayesian policy	$\text{KL}(p(\theta \mid \mathcal{D}, \tau) \parallel p(\theta \mid \mathcal{D}))$ or $I(a; \theta \mid s, \mathcal{D})$ reduction	Direct epistemic information gain when the posterior is well specified.

13 Recommended Reporting Template

For each candidate environment e , report the following after K rollout batches and matched learning updates:

1. action entropy before and after on $\mathcal{S}_{\text{ref}}$;
2. policy KL before versus after on $\mathcal{S}_{\text{ref}}$;
3. probe loss before versus after on a fixed held-out dataset;
4. held-out return or success-rate delta;
5. ensemble or posterior disagreement reduction, if available;
6. posterior KL, if a Bayesian posterior exists.

Then classify the result:

- **Useful information gain:** probe loss improves, held-out behavior improves, and disagreement or posterior entropy decreases.
- **Decisiveness without evidence:** action entropy drops but probe loss and held-out performance do not improve.
- **Surprise without learning:** TD error or rollout loss is high, but subsequent probe improvement is absent.
- **Instability:** policy KL is large, entropy collapses, and held-out metrics worsen.

14 Oracle Evaluation of Approximate Metrics

To compare approximations carefully, define an expensive oracle that estimates the actual utility of training on each candidate environment. The oracle need not be deployable during normal training. Its purpose is to create a reference ranking against which cheap metrics can be judged.

14.1 Notation for the Oracle Experiment

The oracle experiment has three layers: candidate environments, repeated trials on each candidate, and a fixed target evaluation distribution. The notation is:

Symbol	Meaning
$\mathcal{E}_{\text{cand}}$	The pool of candidate environments being ranked now. In a maze setting, these are the candidate levels available for replay or training.
M	Number of candidate environments in the pool.
e_j	The j th candidate environment, with $j \in \{1, \dots, M\}$.
K	Number of independent oracle trials per candidate environment. Larger K gives a lower-variance oracle estimate.
k	Trial index, with $k \in \{1, \dots, K\}$.
θ^-	The same saved pre-learning agent parameters used at the start of every oracle trial.
$\tau_{j,k}$	The rollout batch collected from candidate environment e_j in trial k , using the pre-learning agent.
$U(\theta^-, \tau_{j,k})$	The fixed learning procedure applied to the cloned agent using rollout batch $\tau_{j,k}$.
$\theta_{j,k}^+$	The post-learning parameters after training clone k on candidate environment e_j .
$p_{\text{target}}(e)$	The target evaluation distribution. This should be held fixed and should represent the environments where generalization matters.
$J_{\text{target}}(\theta)$	Evaluation performance of agent θ on the target distribution, such as mean return or success rate.
$O(e_j)$	The expensive oracle score for candidate environment e_j .
$A(e_j)$	A cheap approximation score for candidate environment e_j .

The key design principle is that every candidate starts from the same θ^- . Otherwise, a candidate may look good simply because it was evaluated from a better initial checkpoint. The target distribution must also stay fixed. Otherwise, the score mixes “this environment taught the agent” with “the evaluation distribution changed.”

14.2 Oracle Target

Let $\mathcal{E}_{\text{cand}} = \{e_1, \dots, e_M\}$ be candidate environments. For each e_j , start from the same pre-learning agent state θ^- . Run K independent rollout-and-update trials:

$$\theta_{j,k}^+ = U(\theta^-, \tau_{j,k}), \quad \tau_{j,k} \sim p(\tau \mid e_j, \theta^-). \quad (46)$$

Then evaluate each updated agent on a fixed target distribution $p_{\text{target}}(e)$ or a fixed held-out suite:

$$O_{\text{return}}(e_j) = \frac{1}{K} \sum_{k=1}^K [J_{\text{target}}(\theta_{j,k}^+) - J_{\text{target}}(\theta^-)]. \quad (47)$$

This is a behavioral oracle: it measures actual downstream improvement caused by training on e_j .

If a calibrated Bayesian posterior is available, one can also define a posterior oracle:

$$O_{\text{Bayes}}(e_j) = \frac{1}{K} \sum_{k=1}^K \text{KL} (p(\theta \mid \mathcal{D}, \tau_{j,k}) \parallel p(\theta \mid \mathcal{D})), \quad (48)$$

or the corresponding reduction in posterior-predictive action disagreement on $\mathcal{S}_{\text{ref}}$. This oracle is closer to epistemic information gain, but it depends on the posterior representation being meaningful.

14.3 Approximation Metrics to Compare

For each candidate environment e_j , compute cheap scores before committing to expensive training:

$$A_{\text{entropy}}(e_j) = \Delta H_{\text{action}}(e_j), \quad (49)$$

$$A_{\text{policyKL}}(e_j) = \Delta_{\text{KL}}(e_j), \quad (50)$$

$$A_{\text{probe}}(e_j) = \Delta L_{\text{probe}}(e_j), \quad (51)$$

$$A_{\text{PACE}}(e_j) = \left\| \theta_j^+ - \theta^- \right\|_2^2, \quad (52)$$

$$A_{\text{ens}}(e_j) = U_{\text{epi}}^-(\mathcal{S}_{\text{ref}}) - U_{\text{epi},j}^+(\mathcal{S}_{\text{ref}}), \quad (53)$$

$$A_{\text{modelIG}}(e_j) = \sum_{(s,a) \in \tau_j} \text{I}(\tilde{s}', f^* \mid s, a, \mathcal{D}). \quad (54)$$

These correspond, respectively, to decisiveness, behavioral movement, held-out learning progress, PACE-style parameter change, ensemble epistemic disagreement reduction, and MaxInfoRL/MAX-style model information.

14.4 Worked Toy Example

Consider three candidate maze levels:

- e_1 : an easy maze the agent mostly solves already,
- e_2 : a medium maze with a new useful trap pattern,
- e_3 : a very hard maze that is mostly unsolvable for the current agent.

Suppose the pre-learning agent has target-suite success $J_{\text{target}}(\theta^-) = 50\%$. The oracle clones the same agent three times per candidate, trains each clone on that candidate, and then evaluates on the same held-out target maze suite. For simplicity, write the averaged result directly:

Level	After-training target success	Oracle value $O_{\text{return}}(e_j)$	Interpretation
e_1	52%	+2 percentage points	Slightly useful.
e_2	65%	+15 percentage points	Most useful.
e_3	48%	−2 percentage points	Harmful or too hard.

The oracle ranking is therefore

$$e_2 \succ e_1 \succ e_3. \quad (55)$$

Now compare cheap scores computed before paying for the full oracle:

Approximation	$A(e_1)$	$A(e_2)$	$A(e_3)$	Proxy pick
Action entropy reduction	0.10	0.25	0.60	e_3
TD error magnitude	0.20	0.45	0.90	e_3
Probe loss reduction	0.03	0.18	-0.04	e_2
PACE parameter change	0.05	0.30	0.12	e_2
Ensemble disagreement reduction	0.02	0.22	0.01	e_2

In this toy example, entropy reduction and TD error pick e_3 because the hard maze causes surprise and sharp policy changes. But the oracle shows that training on e_3 hurts target performance. Probe loss reduction, PACE parameter change, and ensemble disagreement reduction pick e_2 , matching the oracle best environment. The conclusion would be that entropy reduction and TD error are detecting disruption or difficulty, while the latter three proxies better track useful learning for this checkpoint and environment family.

14.5 Comparison Criteria

The oracle comparison should not only ask which score gives the best final policy. It should ask whether the approximation tracks the oracle at the level of candidate environments:

- **Rank correlation:** Spearman or Kendall correlation between $A(e_j)$ and $O(e_j)$ across candidate environments.
- **Top- q overlap:** fraction of oracle top environments recovered by the approximation.
- **Selection regret:** oracle value lost by choosing the top approximation-ranked environment instead of the top oracle-ranked environment.
- **Calibration:** whether score magnitude predicts expected downstream improvement, not just rank.
- **Robustness:** whether the correlation survives different seeds, training phases, environment families, and target evaluation distributions.

This design turns the conceptual question into a falsifiable experiment. If action entropy reduction ranks environments poorly against O_{return} , it is merely a decisiveness metric. If PACE or probe-loss reduction tracks O_{return} well, it is a useful single-policy proxy. If ensemble disagreement reduction tracks O_{Bayes} or O_{return} , it is a stronger epistemic proxy. If model information gain tracks only novelty but not downstream improvement, it may be valuable for exploration but insufficient as a curriculum score.

15 Conclusion

The first-principles definition of epistemic information gain is expected posterior entropy reduction over the latent quantity the agent is uncertain about. In RL, action entropy is tempting because it is easy to compute, but it mostly measures decisiveness. Entropy collapse is the failure mode where this decisiveness appears before reliable knowledge.

For a single non-Bayesian policy, epistemic information gain must be approximated by learning-progress metrics such as probe loss reduction, policy-update KL, TD error, and held-out behavioral improvement. For an ensemble, disagreement reduction gives a useful epistemic proxy. For a Bayesian policy, posterior KL or posterior-predictive mutual information gives the cleanest measurement.

Recent work supplies complementary approximations: PACE uses parameter change, MaxInfoRL uses model information gain, MAX uses ensemble disagreement over dynamics, and ITTS uses information-theoretic policy relationships between tasks. The strongest way to compare these approximations is to construct an expensive oracle that measures actual downstream improvement or true posterior change, then evaluate which cheap score best predicts the oracle ranking.

References

- Shannon, C. E. (1948). A mathematical theory of communication.
- Lindley, D. V. (1956). On a measure of the information provided by an experiment.
- MacKay, D. J. C. (1992). Information-based objective functions for active data selection.
- Gal, Y., Islam, R., and Ghahramani, Z. (2017). Deep Bayesian active learning with image data.
- Haarnoja, T., Zhou, A., Abbeel, P., and Levine, S. (2018). Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor.
- Schulman, J., Wolski, F., Dhariwal, P., Radford, A., and Klimov, O. (2017). Proximal policy optimization algorithms.
- Yuan, F., Yin, Q., Shen, S., Xie, Y., Yang, J., Qin, L., Zeng, J., and Li, Q. (2026). PACE: Parameter Change for Unsupervised Environment Design. [arXiv:2605.01358](#).
- Sukhija, B., Coros, S., Krause, A., Abbeel, P., and Sferrazza, C. (2025). MaxInfoRL: Boosting exploration in reinforcement learning through information gain maximization. ICLR. [Project page](#).
- Shyam, P., Jaskowski, W., and Gomez, F. (2019). Model-Based Active Exploration. ICML. [PMLR 97](#).
- Luna Gutierrez, R., and Leonetti, M. (2020). Information-theoretic Task Selection for Meta-Reinforcement Learning. NeurIPS. [arXiv:2011.01054](#).
- Rutherford, A., Beukman, M., Willi, T., Lacerda, B., Hawes, N., and Foerster, J. (2024). No Regrets: Investigating and Improving Regret Approximations for Curriculum Discovery. NeurIPS. [arXiv:2408.15099](#).